



AI Playbook and Toolkit for Public Health Departments

Advanced Model Validation and Evaluation for Public Health AI

ANL 300

Model validation is the process of determining whether an AI tool performs adequately for its intended use, population, setting, data sources, and workflow. Basic validation asks whether the tool works at all. Advanced validation asks whether it works well enough, for whom, under what conditions, with what limitations, and with what monitoring requirements. This module extends foundational validation content by focusing on metrics, subgroup performance, calibration, external validation, workflow validation, error review, drift monitoring, and documentation. Public health agencies need this deeper level of validation when AI outputs may influence investigation, outreach, resource allocation, public communication, or operational prioritization. Learners will develop a validation plan that connects technical performance to public health consequences. The goal is to evaluate not only the model, but the full AI-supported workflow.

Level	intermediate/practitioner
Primary track	Analytics and Modeling Track
Appears in tracks	analytics-modeling

Learning Objectives

- Describe the purpose of advanced model validation in public health AI.
- Interpret sensitivity, specificity, positive predictive value, negative predictive value, precision, recall, F1 score, calibration, and subgroup performance.
- Distinguish internal validation, external validation, and workflow validation.
- Identify when model performance is insufficient for the intended use.
- Design monitoring for drift, degradation, error patterns, and unintended consequences.
- Produce an advanced validation plan for an AI-supported workflow.

Module Overview

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Why This Topic Matters for Public Health AI

Validation matters because public health AI tools can appear credible even when they are not fit for operational use. A polished dashboard, confident generated text, or high overall accuracy score does not prove that the tool is safe, equitable, or useful. Public health staff need to evaluate performance in relation to the specific task and the consequences of error.

Model performance is context-dependent. A tool that performs well in one health system or state may fail in another because of different data sources, coding practices, populations, workflows, or reporting laws. A tool that performs well during development may degrade when disease patterns change or new providers begin reporting. Validation must therefore include local or context-specific testing whenever possible.

Advanced validation also requires attention to equity. Overall performance can hide poor performance for small populations, language groups, rural communities, people with unstable housing, or communities with limited healthcare access. Public health validation should examine subgroup performance and data representativeness before AI outputs influence priority setting or resource allocation.

Public Health Example

A public health agency may test a model that identifies incoming case reports likely to require expedited review. Overall accuracy may be high, but that number alone is not enough. Staff need to know whether the model misses rare but severe conditions, whether it performs differently across facilities, whether it over-flags reports from certain communities, and whether users understand how to act on the output.

The agency should validate the model on recent local data, review errors with epidemiologists, test the workflow with actual users, and monitor override patterns after pilot deployment. If the model performs well technically but creates

too many alerts for available staff, the workflow may still fail. Validation must include operational fit.

Technical Deep Dive

Metric selection should start with the public health consequence. Sensitivity may be prioritized when missing a true event is dangerous. Specificity may be prioritized when false positives create substantial burden or unnecessary intervention. Precision and positive predictive value matter when staff act on positive flags. Calibration matters when scores are interpreted as probabilities.

Confusion matrices are useful because they show true positives, false positives, true negatives, and false negatives. Public health teams should translate each cell into operational meaning. A false negative may mean a delayed investigation. A false positive may mean staff time diverted from other work. A true positive may mean earlier action. A true negative may mean confidence to continue routine processing.

External validation is essential when a model will be used outside its development context. Testing on a new jurisdiction, time period, data source, or population can reveal performance problems that internal validation missed. When external validation is not possible before pilot testing, agencies should begin with limited deployment, close monitoring, and conservative use.

Workflow validation examines whether the AI output is understandable, accessible, timely, actionable, and used as intended. A model can be statistically strong but operationally weak if staff do not trust it, cannot see supporting data, cannot override it, or receive alerts too late. Workflow validation should include usability testing, training, documentation, and feedback loops.

Post-deployment validation should be planned before launch. Agencies should define monitoring indicators, review frequency, escalation triggers, and authority to pause use. Monitoring should include performance drift, subgroup changes, user override rates, incident reports, data feed changes, and public health outcomes where appropriate.

Risks, Failure Modes, and Guardrails

Misleading aggregate performance. Overall metrics may hide poor subgroup performance. Guardrails include subgroup testing and equity review.

Metric mismatch. The selected metric may not match the public health consequence. Guardrails include metric selection tied to intended use and error consequences.

Lack of external validation. Models may fail in new settings. Guardrails include local testing, limited pilots, and monitoring.

Workflow failure. A technically accurate model may be unusable or misused. Guardrails include usability testing, training, source visibility, and override documentation.

Application to AI-Supported Workflows

Advanced validation applies to predictive models, NLP tools, generative summaries, RAG outputs, and agentic workflows. For generative AI, validation may include factuality, completeness, source grounding, omissions, and harmful output review. For RAG, validation should test retrieval quality and citation accuracy. For agentic workflows, validation should include action safety, permission boundaries, and rollback procedures.

The evaluation unit should be the AI-supported workflow, not the model alone. Public health agencies should ask whether the full process improves timeliness, quality, equity, safety, and staff experience without creating unacceptable risk.

Reflection Questions

What decision or action will the AI output support?

Which performance metrics match the consequences of error?

Which subgroups, data sources, or settings require separate review?

What would make performance unacceptable for this use case?

How will the agency monitor performance after deployment?

Practical Exercise

Create an advanced validation plan for one AI-supported workflow. Include intended use, validation dataset, comparison standard, performance metrics, subgroup analysis, error review, workflow testing, user feedback, monitoring indicators, escalation triggers, and documentation requirements.

The plan should clearly explain how validation results will be interpreted and who has authority to approve, limit, pause, or reject the AI use.

Expected Artifact or Evidence

Advanced validation plan

Metric and error-consequence table

Subgroup performance review plan

Post-deployment monitoring schedule

Expected Assignment

- Advanced validation plan
- Metric and error-consequence table
- Subgroup performance review plan
- Post-deployment monitoring schedule

Knowledge Check

- 1. Why can overall accuracy be misleading?
- 2. What is workflow validation?
- 3. Which metric is especially important when missed true events could cause harm?
- 4. What does external validation test?
- 5. What should an advanced validation plan include?

References and Resources

- NIST Artificial Intelligence Risk Management Framework: <https://www.nist.gov/itl/ai-risk-management-framework>
- NIST AI RMF Generative AI Profile: <https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence>
- CDC Evaluation Framework: <https://www.cdc.gov/evaluation/>
- WHO Guidance on Large Multi-Modal Models: <https://www.who.int/publications/i/item/9789240084759>